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Some Basic Concepts of Chemistry

NEET Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	85 questions
Answer key	at the end

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MCQ (Single Correct Answer) 85 questions

Q1. RE-NEET 2026

The numbers 17.0145 and 21.0235 were rounded to three figures after the decimal point. The resulting numbers, respectively, are

(A) 17.015 and 21.024
(B) 17.014 and 21.023
(C) 17.015 and 21.023
(D) 17.014 and 21.024

Open solution

Q2. RE-NEET 2026

The amount of carbon dioxide evolved upon complete combustion of $116g$ of n-butane is

(Given: atomic mass in amu $\mathrm{H} = 1, \mathrm{C} = 12$ and $\mathrm{O} = 16$)

(A) 362 g
(B) 352 g
(C) 322 g
(D) 176 g

Open solution

Q3. NEET 2026

The number of hydrogen atoms present in 5.4 g of urea is:

(Given: Molar mass of urea : $60\ \mathrm{g\ mol^{-1}}$

$\mathrm{N_A}$: 6.022×10^{23} particles $\mathrm{mol^{-1}}$)

(A) 1.084×10^{23}
(B) 1.084×10^{22}
(C) 2.168×10^{22}
(D) 2.168×10^{23}

Open solution

Q4. NEET 2026

When $1\mathrm{dm}^3$ of CO_2 gas is passed over hot coke the volume of gaseous mixture after complete reaction at STP becomes $1.4\mathrm{dm}^3$. The composition of the gaseous mixture at STP is :

(A) $0.8\mathrm{dm}^3$ of $\mathrm{CO}, 0.8\mathrm{dm}^3$ of $\mathrm{CO_2}$
(B) $0.8\mathrm{dm}^3$ of $\mathrm{CO}, 0.6\mathrm{dm}^3$ of $\mathrm{CO_2}$
(C) $0.6\mathrm{dm}^3$ of $\mathrm{CO}, 0.8\mathrm{dm}^3$ of $\mathrm{CO_2}$
(D) $0.6\mathrm{dm}^3$ of $\mathrm{CO}, 0.4\mathrm{dm}^3$ of $\mathrm{CO_2}$

Open solution

Q5. NEET 2025

Among the following, choose the ones with equal number of atoms.

A. 212 g of $\mathrm{Na_2CO_3(\ s)}$ [molar mass $= 106\ \mathrm{g}$]

B. 248 g of $\mathrm{Na_2O(s)}$ [molar mass $= 62\ \mathrm{g}$]

C. 240 g of $\mathrm{NaOH(s)}$ [molar mass $= 40\ \mathrm{g}$]

D. 12 g of $\mathrm{H_2(\ g)}$ [molar mass $= 2\ \mathrm{g}$]

E. 220 g of $\mathrm{CO_2(\ g)}$ [molar mass $= 44\ \mathrm{g}$]

Choose the correct answer from the options given below:

(A) B, C, and D only
(B) B, D, and E only
(C) A, B, and C only
(D) A, B, and D only

Open solution

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Q6. NEET 2025

Dalton's Atomic theory could not explain which of the following?

(A) Law of multiple proportion
(B) Law of gaseous volume
(C) Law of conservation of mass
(D) Law of constant proportion

Open solution

Q7. NEET 2024 (Re-Examination)

The amount of glucose required to prepare $250\ \text{mL}$ of $\frac{\text{M}}{20}$ aqueous solution is :

(Molar mass of glucose : $180\ \text{g mol}^{-1}$)

(A) 2.25 g
(B) 4.5 g
(C) 0.44 g
(D) 1.125 g

Open solution

Q8. NEET 2024 (Re-Examination)

$1.0\ \text{g}$ of H_2 has same number of molecules as in:

(A) $14\ \text{g}$ of N_2
(B) $18\ \text{g}$ of H_2O
(C) $16\ \text{g}$ of CO
(D) $28\ \text{g}$ of N_2

Open solution

Q9. NEET 2024 (Re-Examination)

On complete combustion, 0.3 g of an organic compound gave 0.2 g of CO_2 and 0.1 g of H_2O. The percentage composition of carbon and hydrogen in the compound, respectively is:

(A) 4.07% and 15.02%
(B) 18.18% and 3.70%
(C) 15.02% and 4.07%
(D) 3.70% and 18.18%

Open solution

Q10. NEET 2024

1 gram of sodium hydroxide was treated with $25\ \text{mL}$ of $0.75\ \text{M}\ \text{HCl}$ solution, the mass of sodium hydroxide left unreacted is equal to

(A) 750 mg
(B) 250 mg
(C) Zero mg
(D) 200 mg

Open solution

Q11. NEET 2024

The highest number of helium atoms is in

(A) $4\ \text{mol}$ of helium
(B) 4u of helium
(C) $4\ \text{g}$ of helium
(D) $2.271098\ \text{L}$ of helium at STP

Open solution

Q12. NEET 2024

A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :

(Given atomic masses of A=64 ; B=40 ; C=32 u)

(A) A_2BC_2
(B) ABC_3
(C) AB_2C_2
(D) ABC_4

Open solution

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Q13. NEET 2023 Manipur

The density of 1 M solution of a compound 'X' is 1.25 g mL^{-1}. The correct option for the molality of solution is (Molar mass of compound X = 85 g):

(A) 0.705 m (B) 1.208 m (C) 1.165 m (D) 0.858 m

Open solution

Q14. NEET 2023

The right option for the mass of CO_2 produced by heating $20\ g$ of 20% pure limestone is

(Atomic mass of $Ca = 40$)

$$\left[CaCO_3 \xrightarrow{1200\ K} CaO + CO_2\right]$$

(A) 1.76 g (B) 2.64 g (C) 1.32 g (D) 1.12 g

Open solution

Q15. NEET 2022 Phase 2

The density of the solution is 2.15 g mL^{-1}, then mass of 2.5 mL solution in correct significant figures is :

(A) 53.75 g (B) 5375×10^{-3} g (C) 5.4 g (D) 5.38 g

Open solution

Q16. NEET 2022 Phase 1

In one molal solution that contains 0.5 mole of a solute, there is

(A) 500 mL of solvent (B) 500 g of solvent

(C) 100 mL of solvent (D) 1000 g of solvent

Open solution

Q17. NEET 2022 Phase 1

What mass of 95% pure $CaCO_3$ will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction?

$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + 2H_2O_{(l)}$

[Calculate upto second place of decimal point]

(A) 1.25 g (B) 1.32 g (C) 3.65 g (D) 9.50 g

Open solution

Q18. NEET 2021

An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]

(A) CH_4 (B) CH (C) CH_2 (D) CH_3

Open solution

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Q19. NEET 2020 Phase 2

One mole of carbon atom weighs $12\ \text{g}$, the number of atoms in it is equal to, (Mass of carbon-12 is $1.9926 \times 10^{-23}\ \text{g}$)

(A) 6.022×10^{23} (B) 1.2×10^{23} (C) 6.022×10^{22} (D) 12×10^{22}

Open solution

Q20. NEET 2020 Phase 1

Which one of the following has maximum number of atoms?

(A) 1 g of Mg(s) [Atomic mass of Mg = 24] (B) 1 g of O_2(g) [Atomic mass of O = 16]

(C) 1 g of Li(s) [Atomic mass of Li = 7] (D) 1 g of Ag(s) [Atomic mass of Ag = 108]

Open solution

Q21. NEET 2020 Phase 1

The number of protons, neutrons and electrons in $^{175}_{71}$Lu, respectively, are :

(A) 104, 71 and 71 (B) 71, 71 and 104 (C) 175, 104 and 71 (D) 71, 104 and 71

Open solution

Q22. NEET 2019 (Odisha)

The density of 2 M aqueous solution of NaOH is $1.28\ \text{g/cm}^3$. The molality of the solution is [Given that molecular mass of NaOH $= 40\ \text{g mol}^{-1}$]

(A) 1.32 m (B) 1.20 m (C) 1.56 m (D) 1.67 m

Open solution

Q23. NEET 2019

The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is :

(A) 10 (B) 20 (C) 30 (D) 40

Open solution

Q24. NEET 2018

A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with conc. H_2SO_4 . The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be

(A) 1.4 (B) 3.0 (C) 2.8 (D) 4.4

Open solution

Q25. NEET 2018

In which case is number of molecules of water maximum?

(A) 18 mL of water

(B) 0.18 g of water

(C) 0.00224 L of water vapours at 1 atm and 273 K

(D) 10^{-3} mol of water

Open solution

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Q26. NEET 2017

Which of the following is dependent on temperature?

(A) Molarity
(B) Mole fraction
(C) Weight percentage
(D) Molality

Open solution

Q27. NEET 2016 Phase 2

Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2. When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are

(A) 40, 30
(B) 60, 40
(C) 20, 30
(D) 30, 20

Open solution

Q28. AIPMT 2015

What is the mass of the precipitate formed when 50 mL of 16.9% solution of $AgNO_3$ is mixed with 50 mL of 5.8% NaCl solution ? (Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)

(A) 3.5 g
(B) 7 g
(C) 14 g
(D) 28 g

Open solution

Q29. AIPMT 2015

If Avogadro number N_A, is changed from 6.022×10^{23} mol^{-1} to 6.022×10^{20} mol^{-1}, this would change

(A) the mass of one mole of carbon
(B) the ratio of chemical species to each other in a balanced equation
(C) the ratio of elements to each other in a compound
(D) the definition of mass in units of grams.

Open solution

Q30. AIPMT 2015

The number of water molecules is maximum in

(A) 1.8 gram of water
(B) 18 gram of water
(C) 18 moles of water
(D) 18 molecules of water

Open solution

Q31. AIPMT 2015

What is the mole fraction of the solute in a 1.00 m aqueous solution?

(A) 1.770
(B) 0.0354
(C) 0.0177
(D) 0.177

Open solution

Q32. AIPMT 2015 Cancelled Paper

A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture ?

(A) 16 : 1
(B) 2 : 1
(C) 1 : 4
(D) 4 : 1

Open solution

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Q33. AIPMT 2014

Equal masses of H_2, O_2 and methane have been taken in a container of volume V at temperature 27^oC in identical conditions. The ratio of the volumes of gases $H_2 : O_2$: methane would be

(A) 8 : 16 : 1 (B) 16 : 8 : 1 (C) 16 : 1 : 2 (D) 8 : 1 : 2

Open solution

Q34. AIPMT 2014

When 22.4 liters of $H_{2(g)}$ is mixed with 11.2 litres of $Cl_{2(g)}$, each at S.T.P. the moles of $HCl_{(g)}$ formed is equal to

(A) 1 mol of $HCl_{(g)}$ (B) 2 mol of $HCl_{(g)}$

(C) 0.5 mol of $HCl_{(g)}$ (D) 1.5 mol of $HCl_{(g)}$

Open solution

Q35. AIPMT 2014

1.0 g of magnesium is burnt with 0.56 g O_2 in a closed vessel, Which reactant is left in excess and how much ? (At. wt. Mg = 24, O = 16)

(A) Mg. 0.16 g (B) O_2, 0.16 g (C) Mg, 0.44 g (D) O_2 , 0.28 g

Open solution

Q36. NEET 2013 (Karnataka)

In an experiment it showed that 10 mL of 0.05 M solution of chloride required 10 mL of 0.1 M solution of $AgNO_3$, which of the following will be the formula of the chloride (X stands for the symbol of the element other than chlorine)

(A) X_2Cl_2 (B) XCl_2 (C) XCl_4 (D) X_2Cl

Open solution

Q37. NEET 2013

6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of solution is

(A) 0.001 M (B) 0.1 M (C) 0.02 M (D) 0.01 M

Open solution

Q38. NEET 2013

How many grams of concentrated nitric acid solution should be used to prepare 250 mL of 2.0 M HNO_3? The concentrated acid is 70% HNO_3.

(A) 70.0 g conc. HNO_3 (B) 54.0 g conc. HNO_3

(C) 45.0 g conc. HNO_3 (D) 90.0 g conc. HNO_3

Open solution

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Q39. AIPMT 2011 Mains

Which has the maximum number of molecules among the following ?

(A) 44 g CO_2 (B) 48 g O_3 (C) 8 g H_2 (D) 64 g SO_2

Open solution

Q40. AIPMT 2010 Prelims

The number of atoms in 0.1 mol of a triatomic gas is ($N_A = 6.02 \times 10^{23}$ mol^{-1})

(A) 6.026×10^{22} (B) 1.806×10^{23} (C) 3.600×10^{23} (D) 1.800×10^{22}

Open solution

Q41. AIPMT 2010 Prelims

25.3 g of sodium carbonate, Na_2CO_3 is dissolved in enough water to make 250 mL of solution. If sodium carbonate dissociates completely, molar concentration of sodium ion, Na^+ and carbonate ions, CO_3^{2-} are respectively (Molar mass of Na_2CO_3 = 106 g mol^{-1})

(A) 0.955 M and 1.910 M (B) 1.910 M and 0.955 M
(C) 1.90 M and 1.910 M (D) 0.477 M and 0.477 M

Open solution

Q42. AIPMT 2009

10 g of hydrogen and 64 g of oxygen were filled in a steel vessel and exploded. Amount of water produced in this reaction will be

(A) 3 mol (B) 4 mol (C) 1 mol (D) 2 mol

Open solution

Q43. AIPMT 2008

What volume of oxygen gas (O_2) measuread at 0°C and 1 atm, is needed to burn completely 1 L of propane gas (C_3H_8) measured under the same conditions ?

(A) 5 L (B) 10 L (C) 7 L (D) 6 L

Open solution

Q44. AIPMT 2008

How many moles of lead (II) chloride will be formed from a reaction between 6.5 g of PbO and 3.2 g HCl ?

(A) 0.011 (B) 0.029 (C) 0.044 (D) 0.333

Open solution

Q45. AIPMT 2008

An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C, 38.71% and H, 9.67%. The empirical formula of the compound would be

(A) CHO (B) CH_4O (C) CH_3O (D) CH_2O

Open solution

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Q46. AIPMT 2007

An element, X has the following isotopic composition:
^{200}X : 90% ^{199}X : 8.0% ^{202}X : 2.0%

The weighted average atomic mass of the naturally occurring element X is closest to

(A) 201 amu (B) 202 amu (C) 199 amu (D) 200 amu

Open solution

Q47. AIPMT 2007

Concentrated aqueous sulphuric acid is 98% H_2SO_4 by mass and has a density of 1.80 g mL^{-1}. Volume of acid required to make one litre of 0.1 M H_2SO_4 solution is

(A) 16.65 mL (B) 22.20 mL (C) 5.55 mL (D) 11.10 mL

Open solution

Q48. AIPMT 2005

The mole fraction of the solute in one molal aqueous solution is

(A) 0.009 (B) 0.018 (C) 0.027 (D) 0.036

Open solution

Q49. AIPMT 2004

The maximum number of molecules is present in

(A) 15 L of H_2 gas at STP (B) 5 L of N_2 gas at STP

(C) 0.5 g of H_2 gas (D) 10 g of O_2 gas

Open solution

Q50. AIPMT 2002

Which has maximum molecules?

(A) 7 g N_2 (B) 2 g H_2 (C) 16 g NO_2 (D) 16 g O_2

Open solution

Q51. AIPMT 2002

The percentage of C, H and N in an organic compound are 40%, 13.3% and 46.7% respectively then empirical formula is

(A) $C_3H_{13}N_3$ (B) CH_2N (C) CH_4N (D) CH_6N

Open solution

Q52. AIPMT 2002

2.5 litre of 1 M NaOH solution is mixed with another 3 litre of 0.5 M NaOH solution. Then find out molarity of resultant solution.

(A) 0.80 M (B) 1.0 M (C) 0.73 M (D) 0.50 M

Open solution

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Q53. AIPMT 2001

Percentage of Se in peroxidase anhydrous enzyme is 0.5% by weight (at. wt. = 78.4) then minimum molecular weight of peroxidase anhydrous enzyme is

(A) 1.568×10^4 (B) 1.568×10^3 (C) 15.68 (D) 2.136×10^4

Open solution

Q54. AIPMT 2001

Molarity of liquid HCl, if density of solution is 1.17 g/cc is

(A) 36.5 (B) 18.25 (C) 32.05 (D) 42.10

Open solution

Q55. AIPMT 2001

Specific volume of cylindrical virus particle is 6.02×10^{-2} cc/g whose radius and length are 7 $\mathring{A}$ and 10 $\mathring{A}$ respectively. If $N_A = 6.02 \times 10^{23}$, find molecular weight of virus.

(A) 15.4 kg/mol (B) 1.54×10^4 kg/mol

(C) 3.08×10^4 kg/mol (D) 3.08×10^3 kg/mol

Open solution

Q56. AIPMT 2000

Volume of CO_2 obtained by the complete decomposition of 9.85 g of $BaCO_3$ is

(A) 2.24 L (B) 1.118 L (C) 0.84 L (D) 0.56 L

Open solution

Q57. AIPMT 2000

Oxidation numbers of A, B, C are + 2, +5 and -2 respectively. Possible formula of compound is

(A) $A_2(BC_2)_2$ (B) $A_3(BC_4)_2$ (C) $A_2(BC_3)_2$ (D) $A_3(B_2C)_2$

Open solution

Q58. AIPMT 1999

The number of atoms in $4.25\ g$ of NH_3 is approximately

(A) 1×10^{23} (B) 2×10^{23} (C) 4×10^{23} (D) 6×10^{23}

Open solution

Q59. AIPMT 1999

An organic compound containing C, H and N gave the following analysis C = 40%, H = 13.33% and N = 46.67%. Its empirical formula would be

(A) CHN (B) C_2H_2N (C) CH_4N (D) C_2H_7N

Open solution

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Q60. AIPMT 1999

How many gram of dibasic acid (mol. wt. 200) should be present in $100\ \mathrm{mL}$ of the aqueous solution to give $0.1\ \mathrm{N}$?

(A) $1\ \mathrm{g}$ (B) $2\ \mathrm{g}$ (C) $10\ \mathrm{g}$ (D) $20\ \mathrm{g}$

Open solution

Q61. AIPMT 1998

Given the number, $161\ \mathrm{cm}$, $0.161\ \mathrm{cm}$, $0.0161\ \mathrm{cm}$. The number of significant figures for the three numbers is

(A) 3, 4 and 5, respectively (B) 3, 4 and 4, respectively

(C) 3, 3 and 4, respectively (D) 3, 3 and 3, respectively

Open solution

Q62. AIPMT 1998

Haemoglobin contains 0.33% of iron by weight. The molecular weight of haemoglobin is approximately 67200. The number of iron atoms (at. wt. of Fe is 56) present in one molecule of haemoglobin are

(A) 1 (B) 6 (C) 4 (D) 2

Open solution

Q63. AIPMT 1998

In the reaction

$$4\mathrm{NH_3(g)} + 5\mathrm{O_2(g)} \rightarrow 4\mathrm{NO(g)} + 6\mathrm{H_2O(l)}$$

when 1 mole of ammonia and 1 mole of $\mathrm{O_2}$ are made to react to completion, then

(A) 1.0 mole of $\mathrm{H_2O}$ is produced (B) 1.0 mole of NO will be produced

(C) all the oxygen will be consumed (D) all the ammonia will be consumed

Open solution

Q64. AIPMT 1998

An organic compound containing C, H and N gave the following results on analysis: C = 40%, H = 13.33%, N = 46.67%. Its empirical formula would be

(A) $\mathrm{C_2H_7N_2}$ (B) $\mathrm{CH_5N}$ (C) $\mathrm{CH_4N}$ (D) $\mathrm{C_2H_7N}$

Open solution

Q65. AIPMT 1996

$0.24\ \mathrm{g}$ of a volatile gas, upon vaporisation, gives $45\ \mathrm{mL}$ vapour at NTP. What will be the vapour density of the substance? (Density of $\mathrm{H_2} = 0.089$)

(A) 95.93 (B) 59.93 (C) 95.39 (D) 5.993

Open solution

Q66. AIPMT 1996

The amount of zinc required to produce $224\ \mathrm{mL}$ of $\mathrm{H_2}$ at STP on treatment with dilute $\mathrm{H_2SO_4}$ will be

(A) $65\ \mathrm{g}$ (B) $0.065\ \mathrm{g}$ (C) $0.65\ \mathrm{g}$ (D) $6.5\ \mathrm{g}$

Open solution

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Q67. AIPMT 1996

What is the molarity of H_2SO_4 solution, that has a density $1.84\ g/cc$ at $35\ ^\circ C$ and contains 98% by weight?

(A) $18.4\ M$ (B) $18\ M$ (C) $4.18\ M$ (D) $8.14\ M$

Open solution

Q68. AIPMT 1995

The dimensions of pressure are the same as that of

(A) force per unit volume (B) energy per unit volume
(C) force (D) energy.

Open solution

Q69. AIPMT 1995

The number of moles of oxygen in one litre of air containing 21% oxygen by volume, under standard conditions, is

(A) $0.0093\ mol$ (B) $2.10\ mol$ (C) $0.186\ mol$ (D) $0.21\ mol$

Open solution

Q70. AIPMT 1995

The concentration unit, independent of temperature, would be

(A) normality (B) weight volume percent
(C) molality (D) molarity

Open solution

Q71. AIPMT 1994

The total number of valence electrons in $4.2\ g$ of N_3^- ion is (N_A is the Avogadro's number)

(A) $2.1\ N_A$ (B) $4.2\ N_A$ (C) $1.6\ N_A$ (D) $3.2\ N_A$

Open solution

Q72. AIPMT 1994

How many grams of CH_3OH should be added to water to prepare 150 mL solution of 2 M CH_3OH?

(A) 9.6×10^3 (B) 2.4×10^3 (C) 9.6 (D) 2.4

Open solution

Q73. AIPMT 1993

Which of the following fertilizers has the highest nitrogen percentage?

(A) Ammonium sulphate (B) Calcium cyanamide
(C) Urea (D) Ammonium nitrate

Open solution

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Q74. AIPMT 1992

The concentration unit, independent of temperature, would be

(A) normality
(B) weight volume percent
(C) molality
(D) molarity

Open solution

Q75. AIPMT 1991

A 5 molar solution of H_2SO_4 is diluted from 1 litre to a volume of 10 litres, the normality of the solution will be

(A) 1 N
(B) 0.1 N
(C) 5 N
(D) 0.5 N

Open solution

Q76. AIPMT 1990

The number of gram molecules of oxygen in 6.02×10^{24} CO molecules is

(A) 10 g molecules
(B) 5 g molecules
(C) 1 g molecules
(D) 0.5 g molecules.

Open solution

Q77. AIPMT 1990

Boron has two stable isotopes, $^{10}B\ (19\%)$ and $^{11}B\ (81\%)$. Calculate average at. wt. of boron in the periodic table

(A) 10.8
(B) 10.2
(C) 11.2
(D) 10.0

Open solution

Q78. AIPMT 1990

The molecular weight of O_2 and SO_2 are 32 and 64 respectively. At $15^\circ C$ and $150\ mmHg$ pressure, one litre of O_2 contains 'N' molecules. The number of molecules in two litres of SO_2 under the same conditions of temperature and pressure will be

(A) $N/2$
(B) N
(C) $2N$
(D) $4N$

Open solution

Q79. AIPMT 1989

A metal oxide has the formula Z_2O_3. It can be reduced by hydrogen to give free metal and water. $0.1596\ g$ of the metal oxide requires $6\ mg$ of hydrogen for complete reduction. The atomic weight of the metal is

(A) 27.9
(B) 159.6
(C) 79.8
(D) 55.8

Open solution

Q80. AIPMT 1989

Ratio of C_p and C_V of a gas 'X' is 1.4. The number of atoms of the gas 'X' present in 11.2 litres of it at NTP will be

(A) 6.02×10^{23}
(B) 1.2×10^{23}
(C) 3.01×10^{23}
(D) 2.01×10^{23}

Open solution

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Q81. AIPMT 1989

What is the weight of oxygen required for the complete combustion of $2.8\ \text{kg}$ of ethylene?

(A) $2.8\ \text{kg}$ (B) $6.4\ \text{kg}$ (C) $9.6\ \text{kg}$ (D) $96\ \text{kg}$

Open solution

Q82. AIPMT 1989

The number of oxygen atoms in $4.4\ \text{g}$ of CO_2 is

(A) 1.2×10^{23} (B) 6×10^{22} (C) 6×10^{23} (D) 12×10^{23}

Open solution

Q83. AIPMT 1988

At S.T.P. the density of CCl_4 vapour in g/L will be nearest to

(A) 6.87 (B) 3.42 (C) 10.26 (D) 4.57

Open solution

Q84. AIPMT 1988

One litre hard water contains $12.00\ \text{mg}\ Mg^{2+}$. Milli-equivalents of washing soda required to remove its hardness is

(A) 1 (B) 12.16 (C) 1×10^{-3} (D) 12.16×10^{-3}

Open solution

Q85. AIPMT 1988

1 cc N_2O at NTP contains

(A) $\frac{1.8}{224} \times 10^{22}$ atoms (B) $\frac{6.02}{22400} \times 10^{23}$ molecules

(C) $\frac{1.32}{224} \times 10^{23}$ electrons (D) All the above.

Open solution

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Answer Key

1. **D**	2. **B**	3. **D**	4. **B**	5. **D**
6. **B**	7. **A**	8. **A**	9. **B**	10. **B**
11. **A**	12. **B**	13. **D**	14. **A**	15. **C**
16. **B**	17. **B**	18. **D**	19. **A**	20. **C**
21. **D**	22. **D**	23. **C**	24. **C**	25. **A**
26. **A**	27. **A**	28. **B**	29. **A**	30. **C**
31. **C**	32. **D**	33. **C**	34. **A**	35. **A**
36. **B**	37. **D**	38. **C**	39. **C**	40. **B**
41. **B**	42. **B**	43. **A**	44. **B**	45. **C**
46. **D**	47. **C**	48. **B**	49. **A**	50. **B**
51. **C**	52. **C**	53. **A**	54. **C**	55. **A**
56. **B**	57. **B**	58. **D**	59. **C**	60. **A**
61. **D**	62. **C**	63. **C**	64. **C**	65. **B**
66. **C**	67. **A**	68. **B**	69. **A**	70. **C**
71. **C**	72. **C**	73. **C**	74. **C**	75. **A**
76. **B**	77. **A**	78. **C**	79. **D**	80. **A**
81. **C**	82. **A**	83. **A**	84. **A**	85. **D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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